

Antonio Carmagnani

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Dear Hiring Team at Arup,

I trained and practised as an architect, and the part of the work that has held my attention most consistently is the technical resolution of a building rather than its first image, meaning how it is put together, what it is made of, and what those choices cost over the building's life. That interest is what took me to a Master's Degree in Architecture Computation at The Royal Danish Academy, where I worked on computational geometry, new material systems and the environmental performance of construction. Arup is where that way of working has an institutional home, in a Copenhagen team that keeps materials, facades and fire engineering in one room and has delivered buildings like BLOX and the Camp Adventure tower.

At Isay Weinfield Architects, working on large residential, hotel and mixed-use buildings, I sat across both the design team and the technical development team, which meant that facades came to me at the compositional stage and stayed with me into the build-up. That work ran from what the envelope should be to how it would be made, through material selection, fixing and jointing detail, preliminary structural sizing, and a continuous conversation with the structural and services engineers whose input I reviewed against the design intent before it was documented. Handling a facade end to end that way taught me that an envelope holds together only when its appearance, its build-up and its performance are settled in the same decisions rather than passed between people in sequence. That is the work this role is made of, from concept through detailed design and construction, and it is the reason I am applying to a facade team rather than to another architecture studio.

Material research was at the centre of my master's work at The Royal Danish Academy. I developed a construction technique using blue biomass, seashells and algae bound into a rammed composite, running LCA metrics alongside parametric simulation so that structural performance and environmental impact were pushed in the same iteration rather than traded off afterwards. That work became the material basis for the school's installation at the Venice Architecture Biennale 2025, where I produced the physical panels shown in the seven-metre structure built at the Arsenale with CITA, GXN and 3XN. What the period taught me is that a new material only becomes an engineering proposition once someone has taken it far enough that it has to be built and has to stand up. A team whose remit includes developing new facade technologies and a sustainable approach to envelope design is where that research becomes useful rather than academic.

Arup's Technical Specialist Services team is a growing group of engineers and architects building a facade practice across Scandinavia, on projects where the envelope is where most of the building's difficulty sits. That is where I want to keep working, on buildings whose technical resolution carries as much of the architecture as the design does, and to take the envelope specifically as the place to go deeper.

Best,

Antonio Carmagnani